$$\boxed{\begin{aligned} x_1 \to x_2 &= \overline{x_1} \vee x_2 \\ x_1 \downarrow x_2 &= \overline{x_1 \vee x_2} \end{aligned}}$$

$$A(x_1, x_2, x_3, x_4) = (x_1 \vee x_1\overline{x_2}x_3) \to (\overline{x_3}x_4 \downarrow \overline{x_2}x_1)$$

$$= \overline{(x_1 \vee x_1\overline{x_2}x_3)} \vee \overline{(\overline{x_3}x_4 \vee \overline{x_2}x_1)}$$

$$= \overline{x_1}(\overline{x_1} \vee x_2 \vee \overline{x_3}) \vee ((x_3 \vee \overline{x_4})(x_2 \vee \overline{x_1}))$$

$$= \overline{x_1}\,\overline{x_1} \vee \overline{x_1}x_2 \vee \overline{x_1}\,\overline{x_3} \vee x_3x_2 \vee \overline{x_1}x_3 \vee x_2\overline{x_4} \vee \overline{x_1}\,\overline{x_4}$$

$$= \overline{x_1} \vee \overline{x_1}x_2 \vee \overline{x_1}\,\overline{x_3} \vee x_2x_3 \vee \overline{x_1}x_3 \vee x_2\overline{x_4} \vee \overline{x_1}\,\overline{x_4}$$

$$= x_1 \vee x_2x_3 \vee x_2x_4 \leftarrow$$ POENOSTAVLJENA

a)

| $x_1$ | $x_2$ | $x_3$ | $x_4$ | $f(x)$ | $m$ | $M$ |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 | $m_0$ | $M_{15}$ |
| 0 | 0 | 0 | 1 | 1 | $m_1$ | $M_{14}$ |
| 0 | 0 | 1 | 0 | 1 | $m_2$ | $M_{13}$ |
| 0 | 0 | 1 | 1 | 1 | 3 | $M_{12}$ |
| 0 | 1 | 0 | 0 | 1 | 4 | 11 |
| 0 | 1 | 0 | 1 | 1 | 5 | 10 |
| 0 | 1 | 1 | 0 | 1 | 6 | 9 |
| 0 | 1 | 1 | 1 | 1 | 7 | 8 |
| 1 | 0 | 0 | 0 | 0 | 8 | 7 |
| 1 | 0 | 0 | 1 | 0 | 9 | 6 |
| 1 | 0 | 1 | 0 | 0 | 10 | 5 |
| 1 | 0 | 1 | 1 | 0 | 11 | 4 |
| 1 | 1 | 0 | 0 | 1 | 12 | 3 |
| 1 | 1 | 0 | 1 | 0 | 13 | 2 |
| 1 | 1 | 1 | 0 | 1 | 14 | 1 |
| 1 | 1 | 1 | 1 | 1 | 15 | 0 |

B.1 PDNO $V^4(0,1,2,3,4,5,6,7,12,14,15)$

C.1 PKNO $\&^4(7,6,5,4,2)$

B.2 PDNO $A(x_1x_2x_3x_4) = \bar{x}_1\bar{x}_2\bar{x}_3\bar{x}_4 \vee \bar{x}_1\bar{x}_2\bar{x}_3x_4 \vee \bar{x}_1\bar{x}_2x_3\bar{x}_4$

$\vee\ \bar{x}_1\bar{x}_2x_3x_4 \vee \bar{x}_1x_2\bar{x}_3x_4 \vee \bar{x}_1x_2\bar{x}_3x_4$

$\vee\ \bar{x}_1x_2x_3x_4 \vee x_1x_2\bar{x}_3\bar{x}_4 \vee x_1x_2x_3\bar{x}_4$

$\vee\ x_1x_2x_3x_4$

C.2 PKNO $A(x_1x_2x_3x_4) = (x_1 \vee \bar{x}_2 \vee \bar{x}_3 \vee \bar{x}_4)(x_1 \vee \bar{x}_2 \vee \bar{x}_3 \vee x_4)$

$(x_1 \vee \bar{x}_2 \vee x_3 \vee \bar{x}_4)(x_1 \vee \bar{x}_2 \vee \bar{x}_3 \vee x_4)$

$(x_1 \vee x_2 \vee \bar{x}_3 \vee x_4)$

LAN
VUKUŠIĆ
63190321

DN Q2
6. NOV. 2020